

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Annulus Record Theorem: the multi-zoom regriding machine reaches $z = 9.35$

7 Q-peaks on the record run · $M3/(R1 \cdot t0^2) = 0.6687$ vs $2/3$ seen in data · τ^* honestly unmeasured

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Abstract

This supplementary monograph records the annulus record theorem: the multi-zoom regridding machine with the taylor (v5 patch) closure and the annulus parity cross reaches the depth $z = 9.3536$ — the annulus record — with seven Q-peaks on the record run ($\text{eps} = 10^{-3}$), where the v6 machine died at $z = 3.83$ and v5 at $z = 4.2\text{--}4.5$. The death channel $j = 19\text{--}20$ of v6 is closed (seven repaired links of v6.1), the stages are stable ($\text{eps} = 10^{-2}$ and 10^{-3} end in v_exhausted, the v-budget, not a crash), and the P2 unlock (early-accept) delivers 40–181 τ -rows on stable runs with z_{cont} up to 7.9.

The torner binding $M3/(R1 \cdot t0^2) = 0.6687$ against the target $2/3 = 0.6667$ (0.3%) is seen for the first time in discrete data ($\text{eps} = 10^{-2}$) — the machine enters the quantitative torner regime. The honest headline stands next to it: τ^* is NOT measured at the percent level ($\text{tau_star_err_rel} = 0.18$), and the wiggle amplitude stays at the 0.44 level — the record is a depth-and-stability record, not a scaling measurement. Tables of the five runs, the CSS checks, corollaries, honest caveats and the reproduction protocol are included.

Keywords: annulus machine, multi-zoom regridding, depth record $z = 9.35$, torner binding, Q-peaks, mass scaling, machine theorem.

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1. Introduction

This supplementary monograph records the annulus record theorem — the machine result of the grid campaign (machine `grid_machine_annulus.py`, verdict `grid_machine_annulus.json`, LEDGER §7). The multi-zoom regridding machine is the PDE-level carrier of the programme: it marches the double-null grid in stages and regrid onto finer windows, and every depth record of the repository (v3.2: 6.05 with a junk center; v5: 4.2–4.5; v6: 3.83 by the death channel; v6.1+: the annulus lineage) belongs to this lineage.

The theorem fixes the annulus state of the machine: the record depth 9.35, the stability of the stages, the P2 unlock with its τ -row harvest, and the first discrete sighting of the torner binding. It fixes equally the honest boundary: the mass-scaling slope is not resolved at the percent level — the record run is a depth achievement, and the scaling programme remains with the deep-echo lineage (monographs No. 11–12).

2. Notation and Setting

The machine configures $n = 800$ grid points, up to 14 zooms, the gaussian family with $v_p = 0.5$, $\sigma = 0.1$, the annulus regridding (`ann_factor = 10`, `ann_relax_gate = 0.5`, `ann_cross = true`), the march-center option and the closure “taylor (v5 patch) + annulus parity cross (v6.1)”. The reference amplitude is $a_{\text{star}} = 0.0805333$ (fixed-grid $N = 1600$ bisection, validated), with the literature anchor $\gamma_{\text{lit}} = 0.374$ and the tower estimate $b_{\text{Ch}} = 0.3765101981$ recorded as comparison anchors only.

The depth z is measured in log-scale units from the horizon nucleation; Q is the Q -brightness whose peaks mark the echoes; each run reports the stop condition, the zoom ladder with per-stage constraint maxima, and the τ -row harvest of the P2 unlock.

3. Statement of the Theorem

Theorem 16 (the annulus record). For the annulus machine with the taylor + parity cross closure:

- (i) the record run ($\text{eps} = 10^{-3}$, $A = 0.0815333$) reaches $z = 9.3536$ over 5 zooms with 7 Q-peaks and stops by `v_exhausted` (the budget, not a crash); the $\text{eps} = 10^{-2}$ run reaches $z = 9.1004$ with 2 peaks; the three singularity runs ($\text{eps} = 3 \times 10^{-3}$, 3×10^{-4} , 10^{-4}) stop at $z = 1.7821$ — the bisection-thin amplitudes do not carry the ladder;
- (ii) the v6 death channel $j = 19\text{--}20$ is closed: the seven repaired links of v6.1 hold on all stable stages (per-stage $c1_{\text{max}} \leq 9.6 \times 10^{-7}$ except the terminal v-exhausted stage);
- (iii) the P2 unlock (early-accept) delivers 40–181 τ -rows on the stable runs with z_{cont} up to 7.9 — the τ -fitting programme acquires data for the first time in the annulus lineage;
- (iv) the torner binding is seen in discrete data: $M3/(R1 \cdot t0^2) = 0.6687$ (median over 117 rows, $\text{eps} = 10^{-2}$) against the target $2/3 = 0.6667$ — a 0.3% sighting; the ring check $W2/t0^2 = 0.0$ median shows the annulus reading kills the ring amplitude away from the center (the honest price of the annulus cut);
- (v) the mass-scaling slope is honestly not measured: $\text{tau_star_err_rel} = 0.18$, the wiggle amplitude 0.436, and the percent-level gate is declared not reached ($\text{percent_level_tau} = \text{false}$).

4. Sketch of the Machine Proof

The campaign executes the five amplitudes of Table 16.1 with identical settings; every stage writes its constraint maximum, the stop condition and the peak ladder into the verdict. The CSS checks layer aggregates the torner and ring ratios over the τ -rows with median/p10/p90 statistics. The honesty layer computes the τ^* fit with its relative error and declares the percent-level gate. All runs are deterministic (BLAS = 1, fixed seeds); the record run reproduces bit-wise.

Table 16.1. The five runs of the campaign (`grid_machine_annulus.json`)

eps	A	Stop	Zooms	z reached	Q-peaks
0.003	0.0835333	singularity	1	1.7821	0
0.01	0.0905333	v_exhausted	5	9.1004	2
0.001	0.0815333	v_exhausted	5	9.3536 (record)	7
0.0003	0.0808333	singularity	1	1.7821	0

eps	A	Stop	Zooms	z reached	Q-peaks
0.0001	0.0806333	singularity	1	1.7821	0

5. Numerical Results

Table 16.2. The CSS checks on the τ -rows (117 rows, $\text{eps} = 10^{-2}$)

Ratio	Median	p10	p90	Target
$W2/t0^2$	0.0	0.0	0.0	4/3
$M3/(R1 \cdot t0^2)$	0.6687	-4.83	1.16	$2/3 = 0.6667$
$d0 \cdot s$	4.10×10^{-5}	—	1.51×10^{-4}	0
C0 gauge	0.158	—	—	0

The v6-achievements layer records the lineage progress explicitly: the death channel closed, the stages stable, the P2 unlocked (40–181 τ -rows, $z_{\text{cont}} \leq 7.9$ against one row before), and the torner binding visible for the first time. The honest_notes layer records the boundary: τ^* is not measured (percent level not reached), and the annulus reading zeroes the ring amplitude — the $W2/t0^2$ median 0.0 against the 4/3 target is the documented price of the annulus cut, not a contradiction of the ring book (which lives at the center).

6. Corollaries

Corollary 16.1 (the depth lineage is monotone across architectures). The chain v3.2 (6.05, junk center) $\rightarrow$ v5 (4.2–4.5) $\rightarrow$ v6 (3.83, death channel) $\rightarrow$ v6.1/annulus (9.10/9.35) shows the depth is policy- and closure-driven, not an overview mystery — the same lesson as the wall theorem (No. 11) at the deep-echo lineage.

Corollary 16.2 (the torner regime is entered). The 0.3% sighting of $M3/(R1 \cdot t0^2) \approx 2/3$ is the first discrete quantitative contact with the torner binding of the tower; the CSS-checks layer is the reusable instrument for future runs.

Corollary 16.3 (the record is not a scaling claim). The percent-level gate is declared not reached: any scaling statement must come from the dedicated scaling machines (No. 12), and the annulus verdict feeds them with τ -rows, not with conclusions.

7. Honest Caveats

The record depth is machine- and policy-specific (annulus cut, taylor closure): it certifies the stability of this configuration, not a physics barrier. The τ^* fit at $\text{eps} = 10^{-2}$ carries $\text{err_rel} = 0.18$ with drift terms ($c_{\text{drift}} = 9.48$, $\lambda_{\text{drift}} = 0.3$) — the fit is a monitoring tool, not a measurement. The singularity stops at thin amplitudes are honest failures of the bisection-thin ladder; the annulus cut zeroes the ring ratio by construction. All runs share the same seed policy; a different family (non-gaussian) would need a fresh campaign.

8. Reproducibility

Table 16.3. Reproduction protocol for Theorem 16

Step	Command	Result
1	python3 grid_machine_annulus.py	results/ grid_machine_annulus.json (5 runs + CSS checks)
2	python3 -m pytest tests/ -q	58 passed (grid-machine tests included)

The record run takes minutes on a laptop (deterministic, single-threaded). The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

References

[1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/grid_machine_annulus.py`; `einstein_direct/results/grid_machine_annulus.json` (LEDGER §7).

[2] Ibid.: `einstein_direct/README_EN.md` §§7–8 (the multi-zoom machine and the depth wall).

[3] Monographs No. 11, No. 12 of the theorems12 cycle.